SURJ HOMES: Generative CAD Platform

System Architecture & Technical Specification: Retrieval-Augmented Generation (RAG) & Conditional GAN Synergy

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Executive Summary: In SURJ Homes, architectural design is decoupled into two complementary machine learning layers: **(1) Domain-Informed Semantic Retrieval (RAG)** which resolves *what functional spaces must exist* based on building typology and regional codes, and **(2) Adversarial Spatial Layout Generation (cGAN)** which resolves *where and how those spaces are physically arranged* in 2D coordinate space. This hybrid paradigm eliminates generative hallucinations while ensuring watertight, non-overlapping CAD geometry.

1. The Architectural Challenge: Hallucination vs. Spatial Validity

Modern deep generative models (LLMs and standard Diffusion models) face critical limitations when generating precision architectural assets:

- **Pure Language Models lack coordinate awareness:** While LLMs excel at describing building narratives, they cannot guarantee spatial topologies, resulting in non-orthogonal walls, self-intersecting boundaries, or physically impossible geometries.
- **Pure Vision Models lack domain logic:** Generic computer vision GANs produce pixel rasters rather than semantic BIM polygons. They frequently place ICU beds in office buildings or omit required egress stairs and fire circulation corridors.

The SURJ Homes Solution: A pipelined neuro-symbolic framework where RAG establishes programmatic ground truth, and an adversarial generator (cGAN) enforces geometric constraints.

USER PROMPT *'30x20m 2-story hostel'*	**STAGE 1: RAG ENGINE** • Typology Classifier • Database Program Retrieval • Proportional Sizing Weights	**STAGE 2: cGAN LAYOUT** • Latent Vector + Cond Tensor • Generator Bounding Boxes • Discriminator & AABB Solver	**STAGE 3: CAD SYNTHESIS** • Multi-Story SVGs • Stairwell Riser Alignment • 3D GLB Extrusion

2. Deep Dive: Retrieval-Augmented Generation (RAG) Architecture

The RAG layer operates as the domain expert in the pipeline. It parses unconstrained natural language prompts and anchors them against a structured knowledge base of 25+ architectural typologies (Hostels, Hospitals, Laboratories, Schools, Hotels, Airports, etc.).

- **Typology Disambiguation:** Natural language refers to spaces loosely (e.g., 'dorm', 'paying guest', 'student residence'). RAG maps synonyms down to their canonical building typology using longest-substring token matching.
- **Programmatic Space Expansion:** When an architect specifies a building type without enumerating every utility space, RAG injects mandatory architectural code requirements (e.g., communal washrooms, fire stair cores, security vestibules).
- **Multi-Floor Zoning Differentiation:** Level 1 requires public reception, admin offices, and dining halls; upper floors require dormitory wings or quiet study rooms. RAG partitions functional zones vertically across stories.
- **Architectural Weight Allocation:** Rather than naive uniform grids, RAG assigns space-budget weights (Reading Halls = 2.2, Labs = 1.4, Offices = 0.9, Sanitary = 0.55), ensuring realistic proportional floor area distributions.

3. Deep Dive: Conditional GAN (cGAN) Spatial Packing Architecture

Once RAG establishes the room list and footprint dimensions (W × L), the spatial engine models floor plan generation as a Conditional Generative Adversarial Network (cGAN), similar to HouseGAN/House-GAN++ architectures.

Component	Network Architecture	Role in SURJ Homes Pipeline
Generator (G)	Multi-layer perceptron (Linear 256 → 512 → 1024) with BatchNorm1d and LeakyReLU(0.2).	Takes random latent noise z concatenated with one-hot room labels. Decodes candidate bounding boxes: [x, y, width, length] normalized to [0, 1].
Discriminator (D)	Adversarial discriminator (Linear 512 → 256 → 1) with LeakyReLU(0.2) and Dropout(0.3).	Judges whether candidate layouts conform to human-drawn CAD topologies. Penalizes room intersections, invalid aspect ratios, and disconnected spaces.
Spatial Solver (AABB)	Axis-Aligned Bounding Box (AABB) intersection and slice partitioning engine.	Guarantees 100% boundary compliance and zero wall collisions, resolving any micro-overlaps into clean orthogonal partitions.

4. Synergistic Matrix: RAG vs. GAN Collaboration

The following matrix summarizes why neither machine learning technique is sufficient on its own, and how SURJ Homes achieves production fidelity by coupling them:

Functional Dimension	RAG Alone	cGAN Alone	Combined: SURJ Homes Pipeline
Domain Accuracy	High (Retrieves exact rooms for any typology)	Low (Prone to mislabeling or improper zoning)	**Guaranteed Ground Truth:** Correct room programs injected automatically.
Spatial Geometry	None (Output is purely textual / unmapped)	High (Synthesizes 2D coordinates)	**CAD Coordinates:** Exact [x, y, w, l] boundaries mapped in real-time.
Overlap Prevention	Cannot prevent wall collisions	Penalizes overlaps via discriminator loss	**100% Collision-Free:** Validated by geometric AABB solver pass.
Multi-Story BIM	Understands floor levels semantically	Lacks vertical awareness without conditioning	**Vertical Alignment:** Coordinated stair risers and distinct upper suites.

5. Technical Verification & Future Extensibility

The SURJ Homes platform provides full traceability through interactive UI modals and modular PyTorch codebases (models/gan/, models/vae/, models/diffusion/). The pipeline is production-ready for training on the RPLAN vector floor plan dataset (80,000+ CAD records) on NVIDIA GPUs to replace procedural solvers with fully converged checkpoint weights without breaking frontend UI contracts.